

ROMANOVKA, Russian Federation

WMO: 306500

Lat: 53.200N

Lon: 112.783E

Elev: 923

StdP: 90.72

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.6	-38.1	-43.0	0.1	-40.4	-40.6	0.1	-38.0	8.2	-19.8	6.8	-19.0	0.0	230	0.664

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	27.9	16.4	25.9	15.7	24.0	15.0	18.2	23.9	17.2	22.7	16.4	21.7	2.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	13.2	19.9	15.5	12.3	18.8	14.5	11.5	18.1	55.6	23.6	52.3	22.9	49.5	21.7	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.4	4.7	DB	-43.9	30.9	3.4	2.1	-46.3	32.4	-48.3	33.6	-50.2	34.8	-52.6	36.4	(4)
(5)			WB	-43.5	20.0	3.0	1.1	-45.7	20.8	-47.5	21.5	-49.1	22.1	-51.3	22.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.6	-25.7	-20.7	-11.7	-1.1	6.6	13.9	16.8	13.8	6.5	-2.6	-15.8	-24.3	(6)	
	DBStd	15.78	5.93	5.88	6.58	5.15	3.87	3.61	2.86	3.06	4.09	5.13	7.28	6.60	(7)	
	HDD10.0	5453	1106	860	673	334	119	8	0	6	120	389	774	1062	(8)	
	HDD18.3	8028	1364	1093	932	583	364	138	65	141	354	648	1024	1321	(9)	
	CDD10.0	489	0	0	0	1	13	125	210	125	16	0	0	0	(10)	
	CDD18.3	23	0	0	0	0	0	4	17	2	0	0	0	0	(11)	
	CDH23.3	536	0	0	0	0	9	159	289	72	6	0	0	0	(12)	
	CDH26.7	112	0	0	0	0	1	32	68	11	0	0	0	0	(13)	
	Wind	WSAvg	1.5	1.3	1.4	1.6	2.1	2.2	1.5	1.2	1.2	1.6	1.6	1.4	1.2	(14)
(15) Precipitation	PrecAvg	364	4	3	6	20	35	51	93	87	44	9	8	5	(15)	
	PrecMax	539	11	8	14	44	102	120	183	137	90	30	21	12	(16)	
	PrecMin	237	1	1	2	4	9	7	17	35	9	1	2	0	(17)	
	PrecStd	68	3	2	3	11	19	28	43	29	21	7	5	3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.1	-1.8	8.5	17.9	24.9	29.7	30.9	28.1	23.9	14.8	4.0	-5.4	(19)	
		MCWB	-9.5	-5.0	2.1	7.3	12.8	16.2	17.7	16.2	12.8	7.0	-0.4	-7.0	(20)	
	2%	DB	-11.8	-5.2	5.0	14.0	21.8	27.2	28.6	25.2	20.8	11.2	0.5	-9.0	(21)	
		MCWB	-13.0	-7.7	-0.1	5.5	10.7	15.5	17.0	15.5	11.7	4.6	-2.2	-10.2	(22)	
	5%	DB	-14.2	-7.9	2.2	11.0	19.0	25.0	26.6	23.1	17.6	8.2	-2.1	-11.6	(23)	
		MCWB	-15.0	-9.7	-2.1	3.9	8.9	14.6	16.8	15.0	10.0	2.8	-4.2	-12.5	(24)	
	10%	DB	-16.4	-10.3	-0.5	8.3	16.5	23.0	24.5	21.1	14.9	5.6	-4.6	-14.3	(25)	
		MCWB	-17.0	-11.7	-3.9	2.2	7.8	13.8	16.1	14.7	8.9	1.3	-6.3	-14.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.4	-4.9	2.3	7.7	13.1	18.7	19.8	18.7	13.9	7.2	0.1	-7.0	(27)	
		MCDB	-8.2	-1.9	8.2	17.0	24.3	24.7	25.4	22.6	20.2	14.6	3.5	-5.4	(28)	
	2%	WB	-12.9	-7.6	0.0	5.8	11.1	17.0	18.7	17.2	12.7	4.9	-2.1	-10.1	(29)	
		MCDB	-12.0	-5.3	4.8	13.3	20.3	22.8	24.6	21.3	18.7	10.3	0.5	-9.1	(30)	
	5%	WB	-14.9	-9.7	-2.0	4.2	9.5	15.8	17.8	16.3	11.2	3.2	-4.1	-12.5	(31)	
		MCDB	-14.2	-7.9	2.0	10.6	18.0	22.0	23.8	20.8	16.4	7.7	-2.1	-11.6	(32)	
	10%	WB	-17.0	-11.7	-3.8	2.4	8.2	14.8	17.0	15.4	9.7	1.5	-6.3	-14.9	(33)	
		MCDB	-16.4	-10.3	-0.6	7.8	15.5	21.1	22.8	20.0	13.9	5.3	-4.7	-14.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.6	16.3	16.9	13.3	14.4	14.4	13.1	12.5	13.7	12.6	12.8	11.4	(35)	
		MCDBR	12.1	16.5	17.5	17.9	20.2	19.3	17.5	17.0	19.0	16.5	13.0	10.5	(36)	
		MCWBR	11.2	14.3	12.9	10.8	10.6	8.4	7.3	8.3	10.0	10.4	10.4	9.6	(37)	
	5% WB	MCDBR	12.0	16.4	17.1	17.1	18.8	15.0	13.0	12.9	16.2	15.7	12.9	10.5	(38)	
		MCWBR	11.1	14.2	12.6	10.6	10.1	7.1	6.1	7.0	8.8	10.1	10.3	9.6	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.237	0.264	0.296	0.376	0.480	0.451	0.395	0.367	0.322	0.282	0.258	0.237	(40)	
	taud		2.220	2.201	2.203	2.143	1.900	2.098	2.335	2.394	2.469	2.482	2.307	2.185	(41)	
	Ebn at Noon		760	846	888	850	771	802	849	855	860	830	727	676	(42)	
	Edh at Noon		61	88	110	134	181	151	117	104	85	67	57	53	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.27	2.39	4.05	5.15	5.59	5.99	5.54	4.69	3.62	2.39	1.34	0.94	(44)	
	RadStd		0.04	0.07	0.12	0.31	0.30	0.40	0.51	0.30	0.25	0.16	0.08	0.05	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	-1.62	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page